

### Challenge (Habanero)

- Q1.** A hiker is 3 miles away from the base camp. The absolute value of her distance from the camp is:  
(A)  $|3| = 3$   
(B)  $|3| = -3$   
(C)  $|3| = 0$   
(D)  $|3| = 6$   
(E)  $|3| = 9$
- Q2.** If  $x = -5$ , then  $|x| =$   
(A)  $-5$   
(B)  $5$   
(C)  $0$   
(D)  $10$   
(E)  $-10$
- Q3.** A fuel tank can hold  $|15 - 3|$  gallons of fuel. How many gallons can it hold?  
(A) 8 gallons  
(B) 12 gallons  
(C) 10 gallons  
(D) 15 gallons  
(E) 18 gallons
- Q4.** The absolute value of  $|x - 2|$  is 5 when  $x =$   
(A)  $-3$   
(B)  $7$   
(C)  $-7$   
(D)  $3$   
(E)  $2$
- Q5.** If  $|2x| = 6$ , then  $x =$   
(A)  $pm3$   
(B)  $pm2$   
(C)  $pm1$   
(D)  $3$   
(E)  $-3$
- Q6.** The population of animals in a park is  $|20 - 5|$ . How many animals are in the park?  
(A) 15  
(B) 10  
(C) 20  
(D) 5  
(E) 25
- Q7.** A car travels  $|25 - 10|$  miles in one hour. How many miles does it travel?  
(A) 15 miles  
(B) 20 miles  
(C) 25 miles  
(D) 30 miles  
(E) 35 miles
- Q8.** If  $|x + 2| = 4$ , then  $x =$   
(A)  $pm2$   
(B)  $pm6$   
(C)  $2$   
(D)  $-6$   
(E)  $-2$

#### Open Ended Questions (FRQ)

- Q9.** A group of hikers are planning a trip to a mountain. The trip is  $|2x - 5|$  miles long. If  $x = 4$ , how many miles is the trip? Show your work and explain your answer.
- Q10.** The number of animals in a forest is given by the equation  $|3x - 2|$ , where  $x$  is the number of years since the forest was protected. If  $x = 2$ , how many animals are in the forest? Use the definition of absolute value to explain your answer.

#### Chef's Tips

- Tip 1: Remind students that absolute value represents distance from zero on a number line.
- Tip 2: Encourage students to use real-world examples to understand the concept of absolute value.
- Tip 3: Emphasize the importance of evaluating absolute value expressions carefully.